

SYSTEMATIC LITERATURE REVIEW

Kitchenham Process and PRISMA
2020 Standard

Scientific Writing Course

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Outline

- What is SLR?
- Type of review article
- Kitchenham
- PICOC
- PRISMA 2020

SLR

Systematic Literature Review (SLR)

- Research methodology to **collect, identify, and critically analyze** the available research studies (e.g., articles, conference proceedings, books, dissertations) through a systematic procedure [1].
- The Goal:
 - To identify gaps
 - Provide evidence

[1] D. Pati, L.N. Lorusso, How to write a systematic review of the literature
HERD, 11 (1) (2018), pp. 15–30

SLR vs Traditional Review

SLR

- Follow predefined protocol (e.g. PRISMA)
- Use explicit search strategies.
- Answer specific research question
- Transparent and objective
- Structures results

Traditional review

- No strict protocol
- The search process is often informal and not fully documented
- Provide broad overview of a topic
- Subjective
- No standardized structure usually in a form of narrative discussion

Type of review article

Aspect	SLR	Meta-analysis	Scoping Review
Goal	Answer a specific question	Quantify effect size	Explore research landscape
Data Type	Qualitative + quantitative	Quantitative only	Mostly qualitative
Depth	High	Very high (statistical)	Medium
Breadth	Medium	Low	High
Output	Structured synthesis	Statistical results	Thematic mapping

Kitchenham

- The guideline for performing SLR was introduced by Barbara Kitchenham (2007).
- Kitchenham is suitable for software engineering or complex computation.
- Kitchenham SLR is divided into 3 main processes:



Kitchenham - Planning



- Define objective
- Formulated research questions (RQs) using PICOC
- Define protocol:
 - Search strategy
 - Databases
 - Inclusion/exclusion criteria
 - Quality assessment

PICOC

- P** | **Population:** Who or what is affected?
- I** | **Intervention:** What technology or approach is applied?
- C** | **Comparison:** What is the control or alternative?
- O** | **Outcomes:** What are the measurable effects?
- C** | **Context:** In what specific environment does this take place?

Example PICOC

Automated essay scoring using transformer based

PICOC Element	Description	Example
P (Population)	What is being studied	Student essays / written responses
I (Intervention)	Method used	Transformer-based models (e.g., BERT, RoBERTa, GPT)
C (Comparison)	Alternative (optional)	Traditional ML (SVM), RNN/LSTM, or human scoring
O (Outcome)	What is measured	Accuracy, QWK (Quadratic Weighted Kappa), fairness
C (Context)	Setting/environment	Educational datasets (ASAP, TOEFL), e-learning systems

RQs

Example of RQs derived from PICOC

RQ1: (Intervention)

- What transformer-based models are used for automated essay scoring?

RQ2: (Intervention + Outcome)

- How do transformer-based models perform in essay scoring?

RQ3: (Intervention + Comparison)

- How do transformer-based models compare to traditional approaches?

RQ4: (Outcome)

- What datasets are used in transformer-based AES?

Define protocol - search strategy

- The search terms extracted from the PIC (**Population**, **Intervention**, and **Context**)
- Concept Breakdown
 - **P - Essay**: “automated essay scoring”, “essay scoring”, “automated writing evaluation”, “AES”
 - **I - Transformer model**: “transformer”, “BERT”, “RoBERTa”, “GPT”, “IndoBERT”
 - **C - Education dataset**: “education”, “e-learning”

Define protocol – databse

- Refers to the online repository to search for the articles
- Recommendation:
 - Scopus
 - IEEE Explorer
 - ACM Digital Library
 - ScienceDirect
 - Springer
 - Google Scholar (X)
 - **not recommended as main sources, can be use to find additional papers**

Define protocol – inclusion and exclusion

- Define rules to decide:
 - Which paper should be included
 - Which paper should not be included
- Note: this criterion must be defined before searching
- **Aims:**
 - Prevent bias
 - Ensure consistency
 - Make sure the SLR is reproducible

Example - inclusion and exclusion

Inclusion

- Population: Essay scoring or writing evaluation
- Intervention: Use a transformer-based
- Context: Education
- Quality constraint: peer-reviewed journal/ conference
- Time constraint: published $\geq$ 2021
- Language: English

Exclusion

- Population: Not related to essay scoring
- Intervention: using ML
- Low quality: not peer-reviewed (e.g. from Arxiv)
- Time constraint: published $\leq$ 2021
- Language: Non-English
- Duplicate paper
- Short paper

Define protocol – quality assessment

- Quality assessment is used to evaluate the paper and ensure only high-quality papers are accepted.
- The quality assessment is measured using QA questions
- QA questions can be derived from
 - The PICOC → what matters in the topic (e.g. the method, the evaluation, the dataset)
 - The good research practices → clear method, validity of the experiment, and proper evaluation

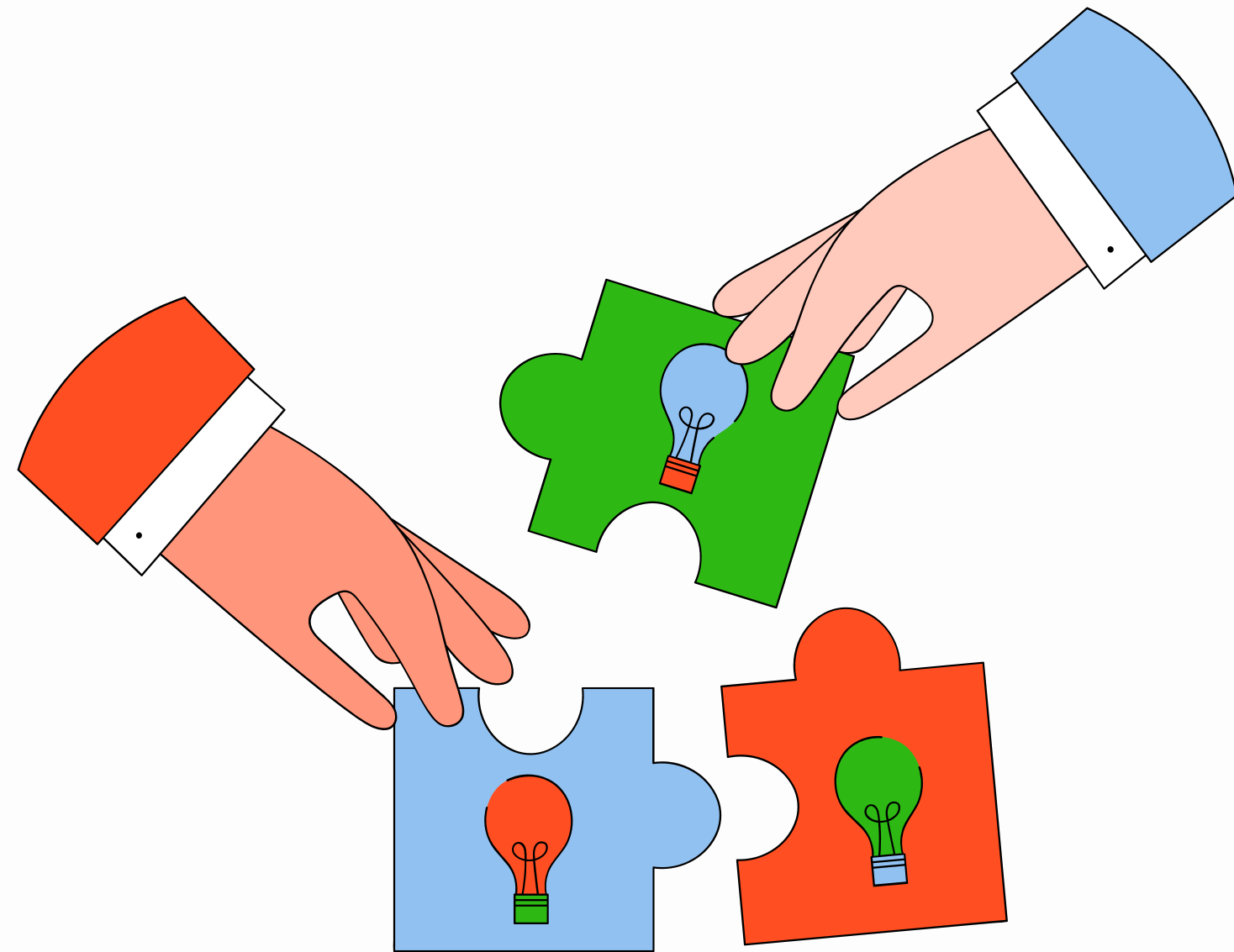
Example of QA question

- QA1: Is the research objective clearly stated?
- QA2: Is the transformer-based model clearly described?
- QA3: Are the model configuration details provided?
- QA4: Does the dataset clearly describe?
- QA5: Are evaluation metrics clearly defined?
- QA6: Is there a comparison with the baseline?

Note:

- *The ideal number of QA questions is 5-10 questions*
- *Use a clear scoring scale (e.g., 1 – 3)*
- *Set the threshold (e.g., $\geq 70\%$ from total maximum score --> include)*

Kitchenham - Conducting



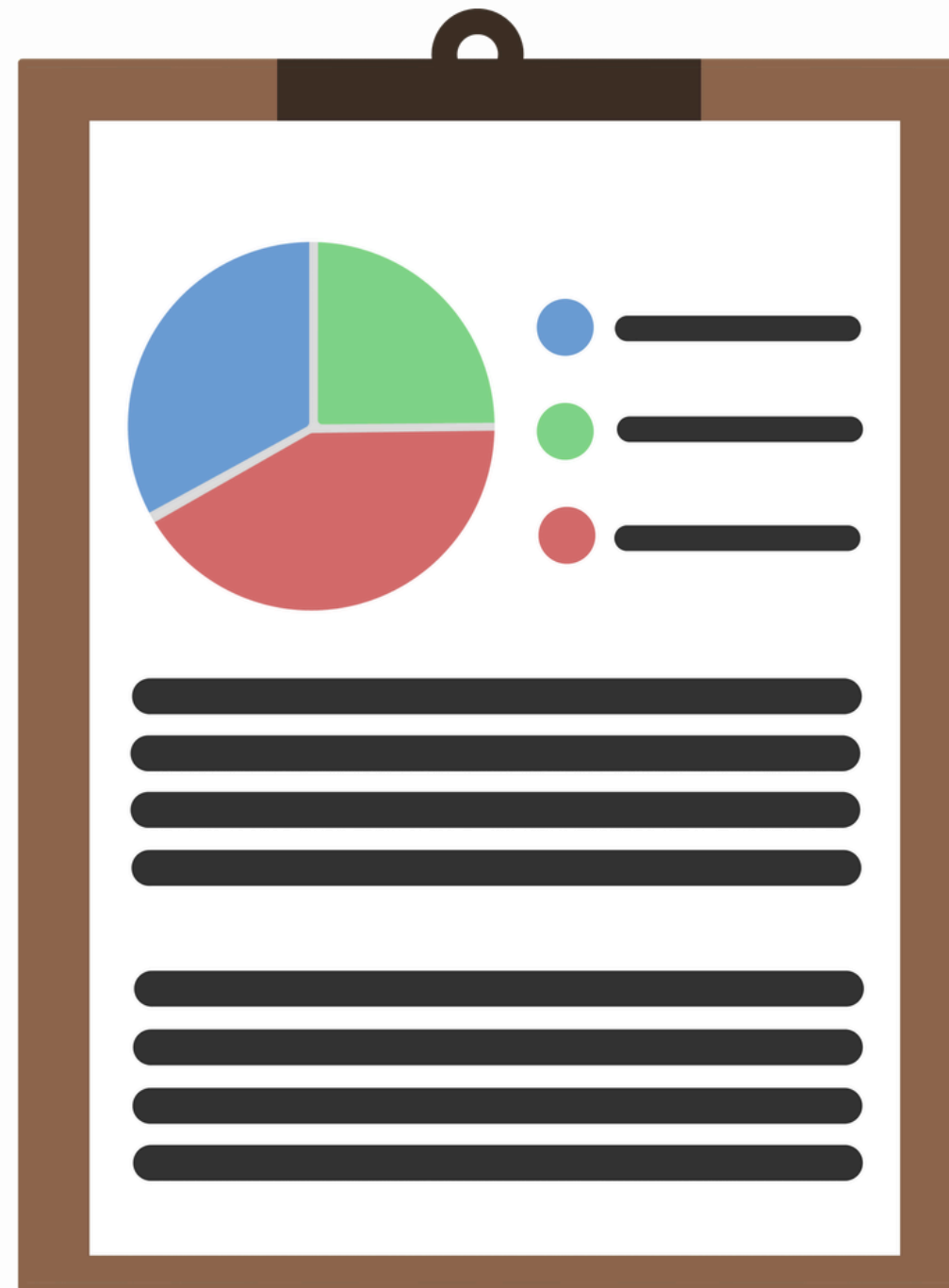
- **Search and Selection**

- Search from the selected database
- Use the key concept from the planning phase
- Apply inclusion and exclusion criteria

- **Quality and Extraction**

- Evaluate the paper based on the QA questions
- Extract information into a table

Kitchenham - Reporting



- The step to present, analyze, and communicate the findings of the SLR in a structured and transparent way

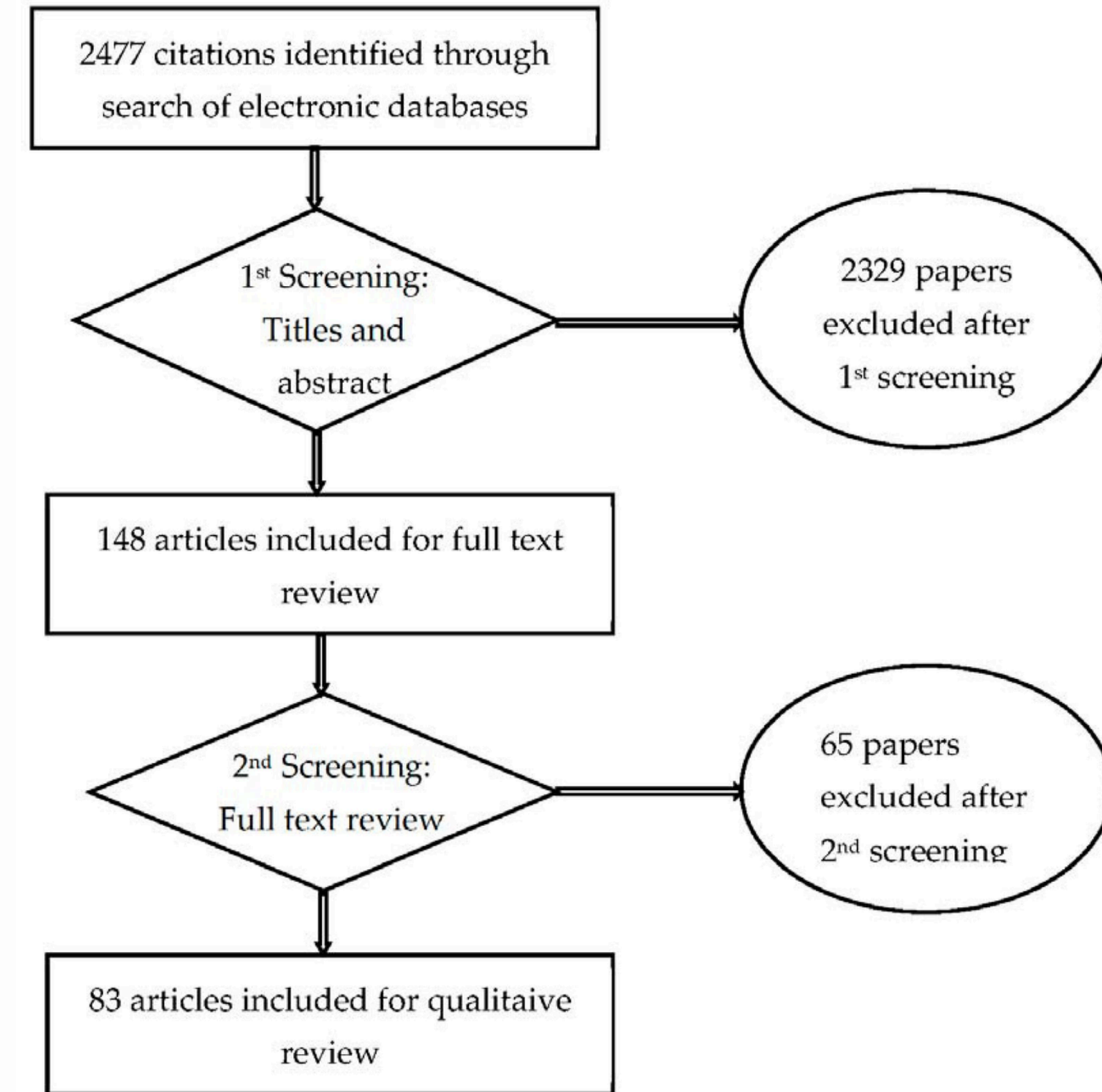
Step-by-step

1. Report study selection (PRISMA)
2. Describe the included studies
3. Present the data extraction result
4. Answer each RQs
5. Synthesize findings
6. Discussion
7. Conclusion

PRISMA Flow example

PRISMA 2020 has 27 checklists to write the report of SLR
<https://www.prisma-statement.org/prisma-2020-checklist>

Aims of PRISMA 2020 to ensure the **transparency** and **reproducibility** of SLR and meet the **global standard**.



The unified blueprint

Kitchenham Planning

- Define PICOC & methods
- PRISMA items 4&5
- Define protocol
- PRISMA item 24

Kitchenham Conducting

- Execute search and selection
- PRISMA items 6-9, 16
- Flow diagram
- Quality assessment & extraction
- PRISMA items 11,18,19

Kitchenham Reporting

- Synthesize data
- PRISMA items 13, 20-22
- Structure Final Manuscript
- PRISMA items 1-3, 23-27

Describe the included studies

- Total number of included studies
- General topic (transformer-based AES)
- Trends (year, growth)
- Key elements:
 - Models
 - Datasets
 - Metrics

Present the data extraction - example

#	Ref	Model	Dataset	Evaluation Metrics	Application
1	[5]	BERT	ASAP	QWK	Fine-tuned R2BERT model using a multi-loss strategy that combined regression and ranking to enhance AES holistic scoring performance.
2	[16]	BERT	ASAP	QWK	Examined if fine-tuning pretrained model affects the scoring performance.
3	[17]	BERT	ASAP (P8), CRP	QWK	Improved BERT performance by employing multiple losses and transfer learning to enhance cross-prompt scoring.
4	[28]	SBERT	ASAP	QWK	Proposed a multi-scale hybrid model that utilized SBERT for sentence vectors and LSTM-MoT for document-scale features to improve essays holistic scoring.
5	[29]	SBERT + LSTM	ASAP, OS	QWK	Used a multi-task model that utilized SBERT for essay features extraction and LSTM for sentence processing to generate overall essay and trait scores.
6	[36]	BERT	ASAP	QWK	Suggested combining manual features with deep encoded features to improve AES holistic scoring performance.
7	[37]	BERT / Longformer	ASAP	QWK	Used BERT and Longformer models to improve AES scoring by evaluating the grammatical and stylistic traits to assess the overall essay quality.
8	[42]	BERT	ASAP	QWK	Fine-tuned BERT using three Active Learning strategies to reduce the number of manually scored essays required to build accurate AES systems.
9	[45]	GloVe / ELMo / BERT	ASAP	QWK	Compared various pretrained models' robustness against adversarial essays.

Answer each RQs

- State the RQ clearly
- Present findings from the data
- Support with evidence (**very important**)
- Add a table or figure to support the finding
- Short interpretation

Important principles

- Always use evidence, avoid abstract statements
 - BERT is popular (too abstract and subjective)
 - BERT is used in 70% of studies ← use evidence
- Avoid adding subjective statements
- One RQ for one subsection
 - Common mistake mixing all RQs together
- Brief interpretation → , deep interpretation located in the discussion section

Synthesize findings

- After answering RQs
- Connecting the findings after answering the RQs into a bigger picture
- Perform cross-analysis (**important**)
 - Combine multiple dimensions
 - Example: model + performance + dataset

Discussion

- Deep interpretation of the findings
 - not just showing the statistics or table
 - explain the reason behind it
- Comparison with existing studies
 - Does the finding align with the previous studies
 - Is there any contradiction with previous studies
- Implications of the finding
- Limitation
- Future direction

Assignment

- Create a group of four
- Choose the topic
- Create the Kitchenham → **Planning phase only**
- Submit before the next meeting